MYCOPHENOLATE

Highly teratogenic

Two forms of contraception

CHEMICAL NAME 2-morpholinoethyl (E)-6-(4-hydroxy-6-methoxy-7-methyl-3-oxo-1,3-dihydroisobenzofuran-5-yl)-4-methylhex-4-enoate

CLASS Antiproliferative immunosuppressant (IMPDH inhibitor)

TRADE NAMES CellCept · Myfortic (mycophenolic acid) · Mycophenolate (generic)

THERAPEUTIC USES / INDICATIONS

- Prevention of transplant rejection (with calcineurin inhibitor and steroids).
- Lupus nephritis (induction and maintenance).
- Vasculitis, autoimmune hepatitis, myasthenia gravis, interstitial lung disease (off-label).

PHARMACODYNAMICS

- Inhibits **inosine monophosphate dehydrogenase (IMPDH)** → ↓ guanine nucleotide synthesis → selectively blocks T and B lymphocyte proliferation (lymphocytes rely on de novo purine synthesis).

ADMINISTRATION

- **PO** twice daily on an empty stomach (or with food if GI upset).
- **IV** infusion over ≥2 h.
- Do not crush tablets/open capsules (teratogenic dust); pregnant staff avoid handling.
- MMF 1 g ≈ mycophenolic acid (Myfortic) 720 mg — not interchangeable mg-for-mg.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Diarrhoea, nausea, vomiting, abdominal pain.
- Leucopenia.
- Tremor, headache.

ADVERSE EFFECTS (SERIOUS)

- Teratogenicity (miscarriage, facial/ear malformations).
- Bone-marrow suppression; pure red cell aplasia.
- Serious infections (CMV, PML, BK virus).
- Lymphoma and skin cancer; GI bleeding/perforation.

DOSAGES

- **Renal transplant:** 1 g twice daily (MMF).
- **Liver/heart transplant:** 1–1.5 g twice daily.
- **Lupus nephritis:** induction 1–1.5 g twice daily; maintenance 0.5–1 g twice daily.
- **Children:** 600 mg/m² twice daily.

CONTRAINDICATIONS & CAUTIONS

- Pregnancy; women of childbearing potential not using highly effective contraception; breastfeeding.
- Hypersensitivity.
- Live vaccines.

KEY INTERACTIONS

- **Antacids, cholestyramine, PPIs (MMF):** ↓ absorption.
- **Aciclovir, ganciclovir:** ↑ levels of both.
- **Rifampicin:** ↓ MPA.
- **Azathioprine:** avoid combination.

PHARMACOKINETICS

- **Peak:** 1 h. Prodrug hydrolysed to active mycophenolic acid (MPA); enterohepatic recirculation.
- Glucuronidated in liver.
- Excreted in urine. **t½:** ~17 h.

NURSING MANAGEMENT

- Pregnancy test before starting and 8–10 days later; contraception counselling (2 methods).
- FBC weekly for first month, fortnightly months 2–3, then monthly to 1 year.
- Monitor GI symptoms, infection signs.
- Do not crush; safe handling.

PATIENT EDUCATION

- Use two reliable methods of contraception during and 6 weeks after (men: condoms during and 90 days after).
- Report diarrhoea, fever, sore throat, bruising, unusual tiredness.
- Use sunscreen; avoid live vaccines.
- Do not open or crush capsules/tablets.

KEY POINTS

- IMPDH inhibitor; lymphocyte-selective.
- Highly teratogenic.
- Transplant 1 g BD; lupus nephritis.
- Diarrhoea, marrow suppression, infections.